

Nivell 1

Exercici 1

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.

```
1      -- Creamos la base de datos
2 •    CREATE DATABASE IF NOT EXISTS transactions;
3 •    USE transactions;
4      -- Creamos la tabla company
5 •    CREATE TABLE IF NOT EXISTS company (
6      id VARCHAR(15) PRIMARY KEY,company_name VARCHAR(255),
7      phone VARCHAR(15),email VARCHAR(100),
8      country VARCHAR(100),website VARCHAR(255)
9      );
10     -- Creamos la tabla transaction
11 •    CREATE TABLE IF NOT EXISTS transaction (
12     id VARCHAR(255) PRIMARY KEY,
13     credit_card_id VARCHAR(15) REFERENCES credit_card(id),
14     company_id VARCHAR(20),user_id INT REFERENCES user(id),
15     lat FLOAT,longitude FLOAT,
16     timestamp TIMESTAMP,Amount DECIMAL(10, 2),
17     declined BOOLEAN,
18     FOREIGN KEY (company_id) REFERENCES company(id)
19     );
20
```

Output

Action Output

#	Time	Action	Message
✓ 1	12:57:20	USE transactions	0 row(s) affected
✓ 2	12:57:33	CREATE TABLE IF NOT EXISTS company (id VARCHAR(15) PRIMARY KEY, ...	0 row(s) affected
✓ 3	12:57:36	CREATE TABLE IF NOT EXISTS transaction (id VARCHAR(255) PRIMARY KE...	0 row(s) affected

```

1  -- 100 inserts de company
2  • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2222', 'Ac Fermentum Inco
3  • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2226', 'Magna A Neque Indu
4  • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2230', 'Fusce Corp.', '08
5  • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2234', 'Convallis In Inco
6  • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2238', 'Ante Iaculis Nec f
7  • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2242', 'Donec Ltd', '01 25
8  • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2246', 'Sed Nunc Ltd', '05
9  • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2250', 'Amet Nulla Donec (
10 • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2254', 'Nascetur Ridiculus
11 • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2258', 'Vestibulum Lorem f
12 • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2262', 'Gravida Sagittis l
13 • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2266', 'Mus Aenean Eget Fe
14 • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2270', 'Dis Parturient Ins
15 • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2274', 'Sed LLC', '01 63 1
16 • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2278', 'Arcu LLP', '06 46
17 • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2282', 'Pretium Neque Corq
18 • INSERT INTO company (id, company_name, phone, email, country, website) VALUES (      'b-2286', 'Fringilla LLC', '0

```

Output

Action Output

#	Time	Action	Message
✓ 97	13:09:28	INSERT INTO company (id, company_name, phone, email, country, website) VALUES (...	1 row(s) affected
✓ 98	13:09:28	INSERT INTO company (id, company_name, phone, email, country, website) VALUES (...	1 row(s) affected
✓ 99	13:09:28	INSERT INTO company (id, company_name, phone, email, country, website) VALUES (...	1 row(s) affected
✓ 100	13:09:28	INSERT INTO company (id, company_name, phone, email, country, website) VALUES (...	1 row(s) affected

```

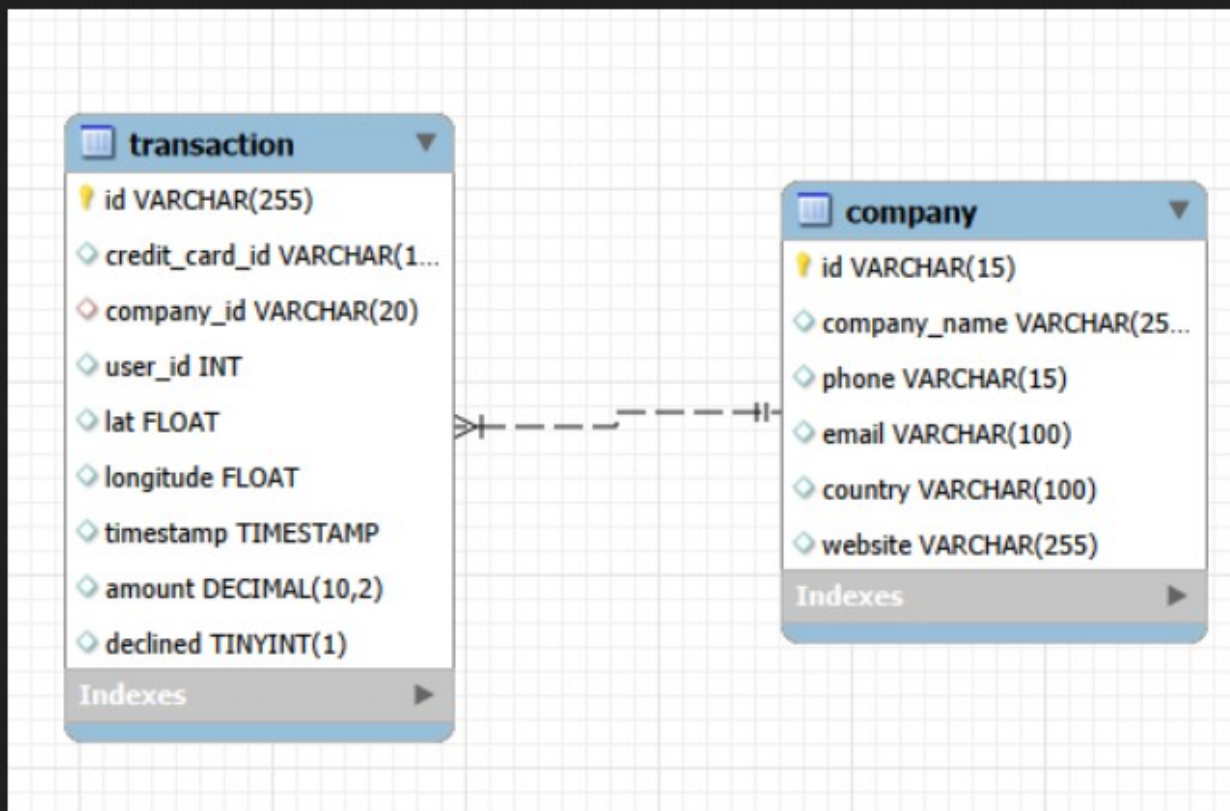
1  • USE transactions; -- 100000 trasnactions total
2  -- Insertamos datos de transaction
3  • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('CDDA7E
4  • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('094563
5  • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('C47C7C
6  • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('2F852E
7  • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('088774
8  • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('A08D09
9  • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('29322E
10 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('3F614E
11 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('133534
12 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('185D9E
13 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('F080DC
14 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('AB0D4E
15 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('ED8865
16 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('44CD5F
17 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('8EA91C
18 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined) VALUES ('C44DE8

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
---	------	--------	---------	------------------



Transaction es la tabla de hechos, company es una tabla de dimensión.

Transaction almacena las transacciones realizadas:

id: identificador único de la transacción (clave primaria).

company_id: clave foránea relacionada con la PK de company.

credit_card_id, user_id: candidatas a FK de las tablas credit card y users.

lat, longitude: ubicación de la transacción. timestamp: fecha y hora. amount: importe de la operación. declined: indica si la transacción fue rechazada (1) o aceptada (0).

Company almacena la información de las empresas:

id: identificador único de la empresa (clave primaria).

company_name, phone, email, country, website: datos descriptivos y de contacto.

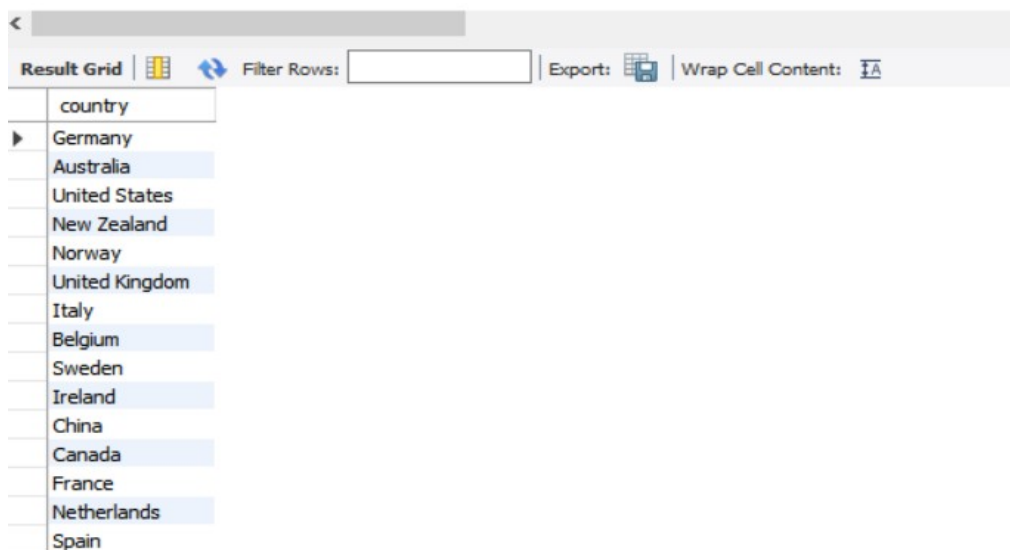
Relación entre las tablas La relación se establece mediante `transaction.company_id` -> `company.id`. Esto significa que: Una empresa puede tener muchas transacciones. Cada transacción pertenece a una sola empresa. Es una relación 1:N (uno a muchos) entre `company` y `transaction`.

Exercici 2

Utilitzant JOIN realitzaràs les següents consultes:

Exercici 2.1 - Llistat de països que generen vendes

```
1 • select DISTINCT c.country from company as c
2   join transaction as t on c.id = t.company_id;
```



country
Germany
Australia
United States
New Zealand
Norway
United Kingdom
Italy
Belgium
Sweden
Ireland
China
Canada
France
Netherlands
Spain



#	Time	Action	Message
1	10:36:46	select DISTINCT c.country from company as c join...	15 row(s) returned

- Exercici 2.2 - Des de quants països es generen les vendes.

```
1 • select count(distinct c.country) from company as c
2   join transaction as t on c.id = t.company_id;
```

Result Grid

count(distinct c.country)
15

Result 1 x

Output

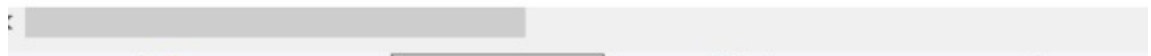
Action Output

#	Time	Action	Message
1	10:36:46	select DISTINCT c.country from company as c join...	15 row(s) returned

Ex 2.3 - Identifica la companyia amb la mitjana més gran de vendes. Usar Limit 1 si se quieren ocultar el resto



```
1 • SELECT c.company_name, ROUND(AVG(t.amount), 2)
2 FROM company AS c
3 JOIN transaction AS t ON c.id = t.company_id
4 GROUP BY t.company_id
5 ORDER BY AVG(t.amount) DESC
6 LIMIT 1;
```



Result Grid |   Filter Rows: | Export:  | Wrap Cell Content:  | Fetch rows

	company_name	ROUND(AVG(t.amount), 2)
▶	Ac Fermentum Incorporated	284.87

Result 2 x

Output

 Action Output ▼

#	Time	Action	Message
1	10:42:35	SELECT c.company_name, ROUND(AVG(t.amoun...	1 row(s) returned

Exercici 3

Ex 3.1. Utilitzant només subconsultes (sense utilitzar JOIN):

- Mostra totes les transaccions realitzades per empreses d'Alemanya.

The screenshot shows a SQL IDE interface. At the top, there's a toolbar with various icons and a dropdown menu set to "Limit to 50000 rows". Below the toolbar, a SQL query is entered in a text area:

```
1  -- Exercici 3.1
2  --      Mostra totes les transaccions realitzades per empreses
3  •  SELECT * from transaction
4     where company_id in
5         (SELECT id from company where country = "Germany")
6     ; -- '13291 transacciones'
```

Below the query editor, there's a "Result Grid" section. It has a toolbar with icons for "Filter Rows", "Edit", and "Export/Import". The grid displays the following data:

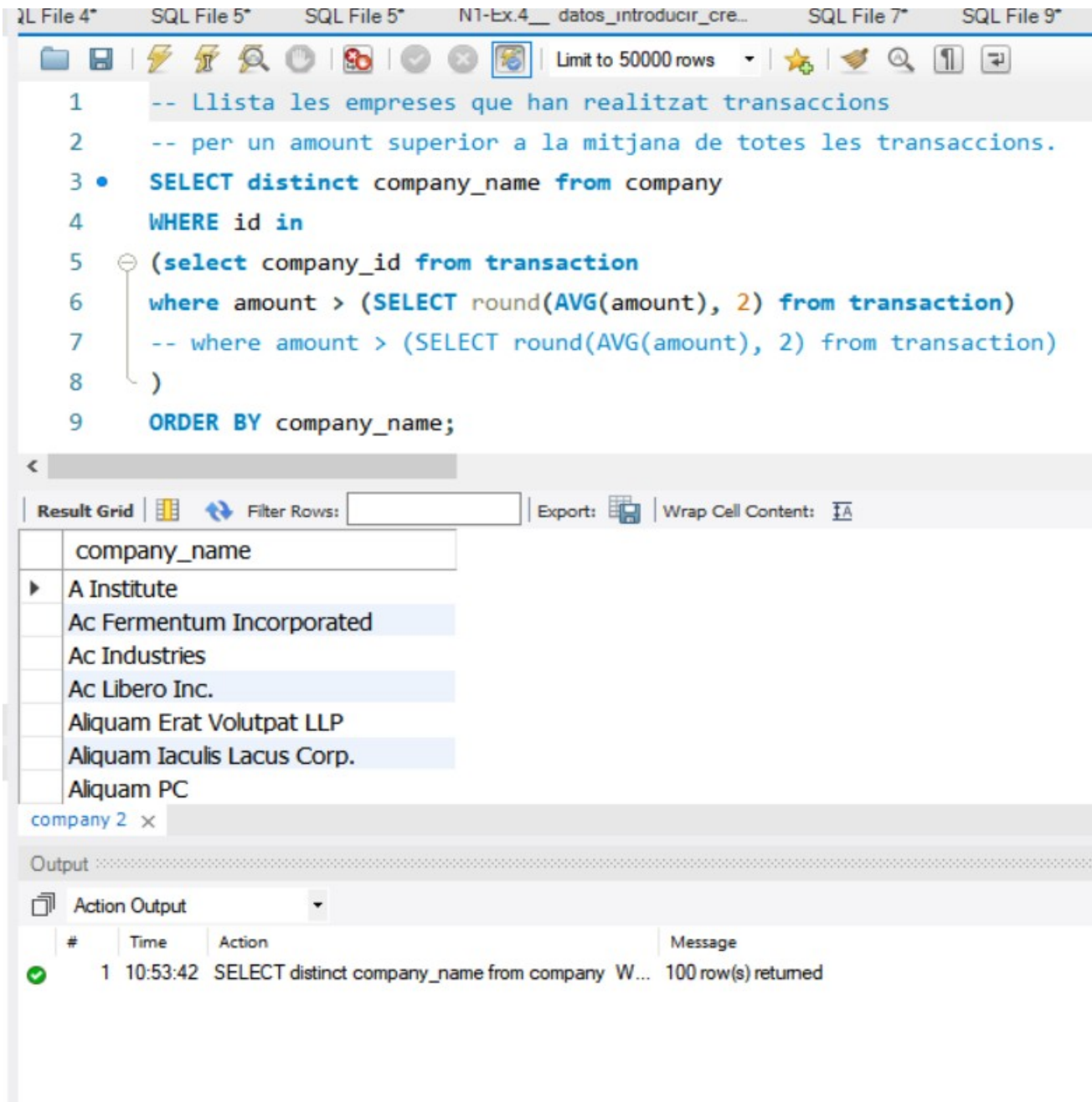
	id	credit_card_id	company_id	user_id	lat
▶	00138D3B-206D-4C03-94B7-63A26...	CcS-4899	b-2222	318	41.
	0013C1B6-3B84-4D6C-8154-E2B3F...	CcS-5070	b-2222	489	41.
	00201A11-2E62-44C4-941D-198FC...	CcU-3512	b-2222	193	55.

Below the result grid, there's an "Output" section. It has a dropdown menu set to "Action Output". The output shows a single message:

#	Time	Action	Message
1	10:51:42	SELECT * from transaction where company_id in ...	13291 row(s) returned

Exercici 3.2

Ex 3.2 Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de les transaccions.



The screenshot shows a SQL IDE with multiple tabs. The active tab is 'N1-Ex.4__ datos_introducir_cre...'. The query editor contains the following SQL code:

```
1  -- llista les empreses que han realitzat transaccions
2  -- per un amount superior a la mitjana de totes les transaccions.
3  • SELECT distinct company_name from company
4    WHERE id in
5    (select company_id from transaction
6     where amount > (SELECT round(AVG(amount), 2) from transaction)
7     -- where amount > (SELECT round(AVG(amount), 2) from transaction)
8    )
9    ORDER BY company_name;
```

Below the query editor, the 'Result Grid' is visible. It shows a table with the following data:

company_name
A Institute
Ac Fermentum Incorporated
Ac Industries
Ac Libero Inc.
Aliquam Erat Volutpat LLP
Aliquam Iaculis Lacus Corp.
Aliquam PC

At the bottom, the 'Output' section shows the 'Action Output' for the query execution:

#	Time	Action	Message
1	10:53:42	SELECT distinct company_name from company W...	100 row(s) returned

Exercici 3.3

Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

The screenshot shows a database management interface. At the top, there is a toolbar with various icons and a dropdown menu set to "Limit to 50000 rows". Below the toolbar, a SQL query is displayed in a text editor:

```
1  -- 3.3  Eliminaran del sistema les empreses que no tenen transaccions registrades,
2  -- entrega el llistat d'aquestes empreses.
3  •  select company_name from company
4  where id not in (
5      select DISTINCT company_id from transaction
6  ) ORDER BY company_name
7  ; -- no hay ninguna sin transacciones
```

Below the query editor, there is a "Result Grid" section. It includes a "Filter Rows" input field, an "Export" button, and a "Wrap Cell Content" checkbox. The "Result Grid" itself is empty, showing only the column header "company_name". To the right of the "Result Grid" is a "Form Editor" button.

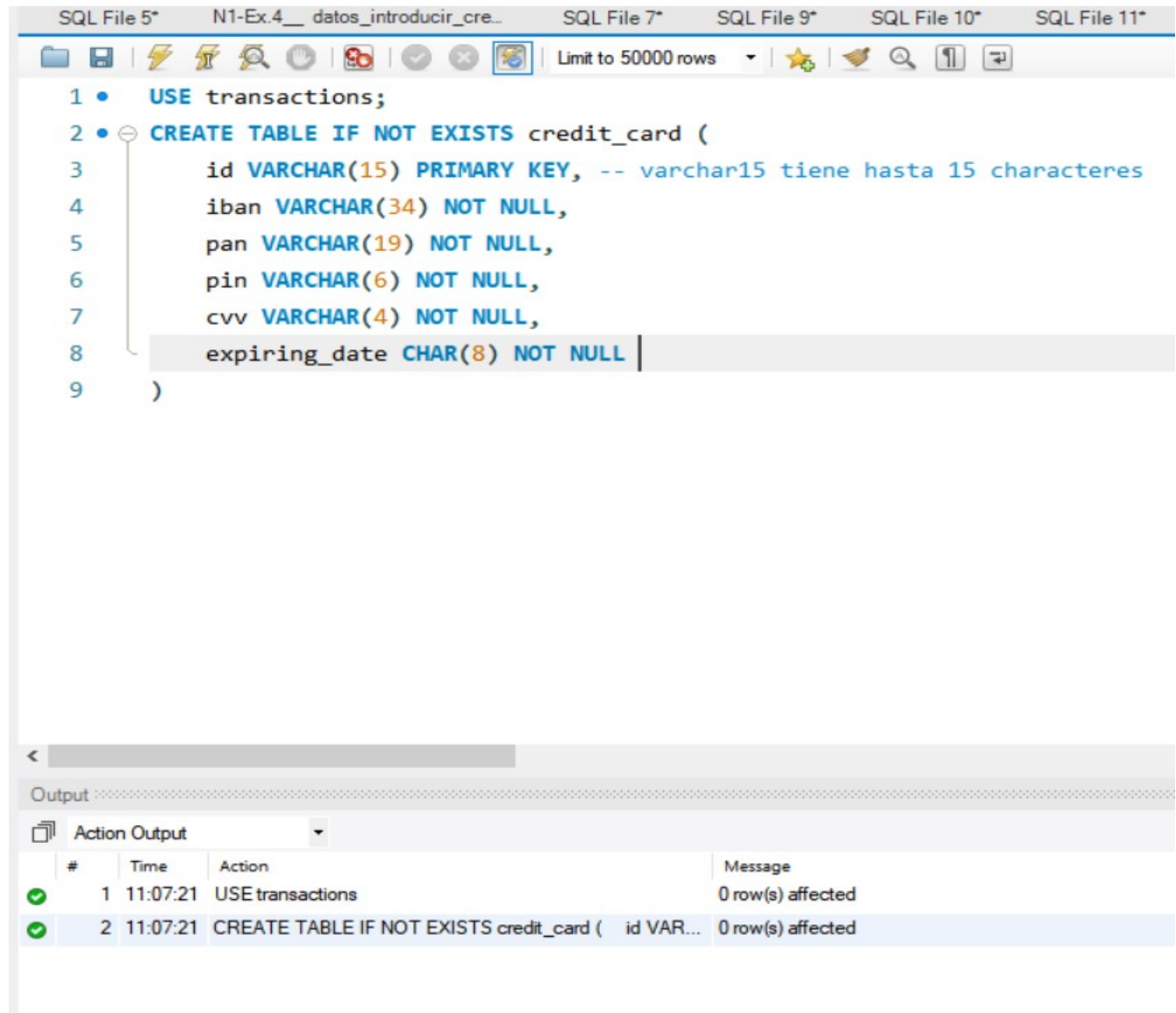
At the bottom, there is an "Output" section. It includes a tab labeled "company 1" and a "Read On" button. Below the "Output" section is an "Action Output" dropdown menu. The "Action Output" table shows the following data:

#	Time	Action	Message	Duration / Fetch
✓ 1	10:57:28	select company_name from company where id not in (...	0 row(s) returned	0.063 sec / 0.000 se

Exercici 4

La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit.

La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company").



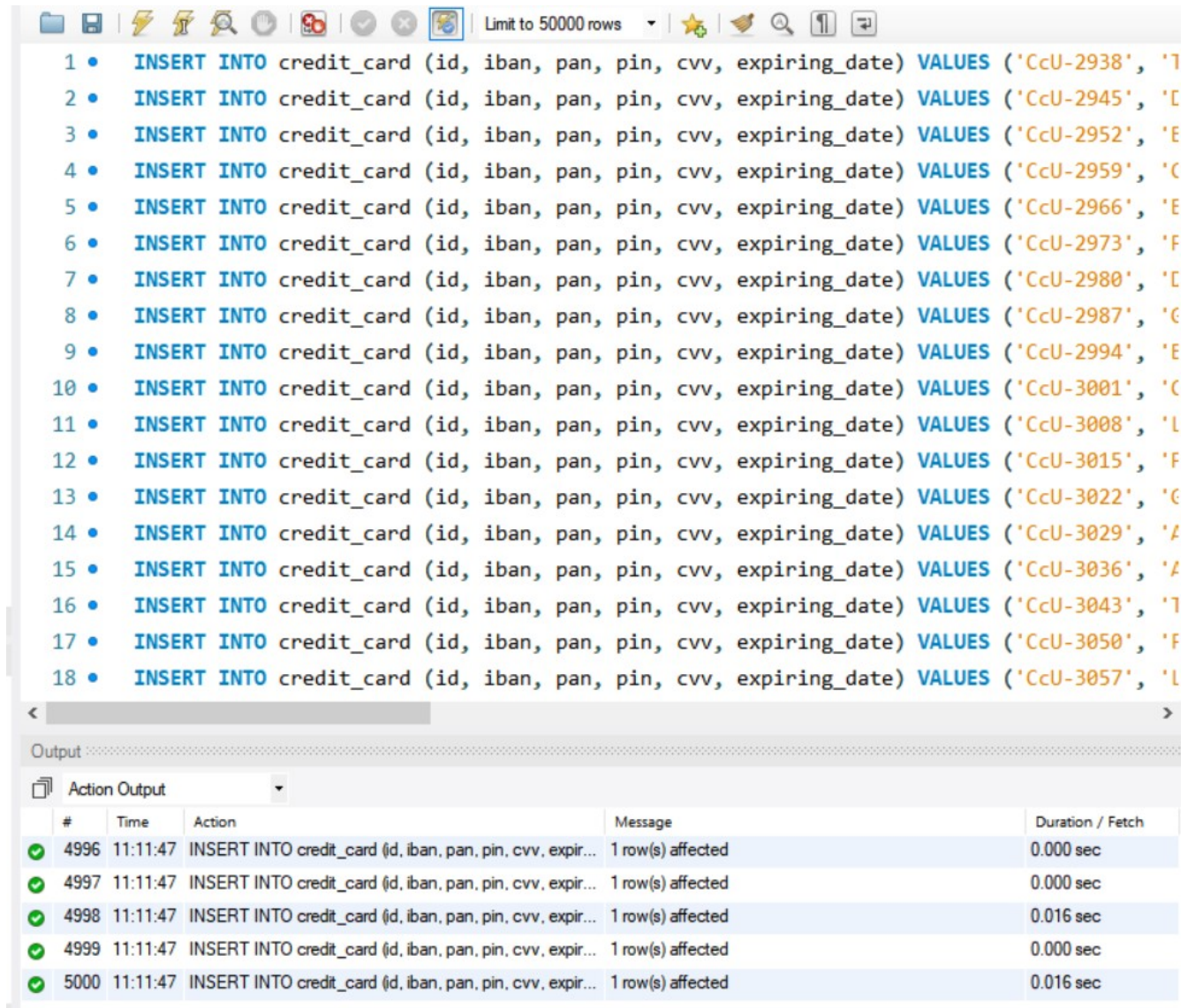
The screenshot shows a SQL IDE with multiple tabs. The active tab is 'N1-Ex.4__ datos_introducir_cre...'. The SQL editor contains the following code:

```
1 • USE transactions;
2 • CREATE TABLE IF NOT EXISTS credit_card (
3     id VARCHAR(15) PRIMARY KEY, -- varchar15 tiene hasta 15 caracteres
4     iban VARCHAR(34) NOT NULL,
5     pan VARCHAR(19) NOT NULL,
6     pin VARCHAR(6) NOT NULL,
7     cvv VARCHAR(4) NOT NULL,
8     expiring_date CHAR(8) NOT NULL
9 )
```

Below the editor is the 'Output' panel, which shows the execution results of the SQL statements:

#	Time	Action	Message
✓ 1	11:07:21	USE transactions	0 row(s) affected
✓ 2	11:07:21	CREATE TABLE IF NOT EXISTS credit_card (id VAR...	0 row(s) affected

-Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit".



The screenshot displays a database management interface. The top section shows a list of 18 SQL `INSERT INTO` statements for a table named `credit_card`. Each statement includes columns `id`, `iban`, `pan`, `pin`, `cvv`, and `expiring_date`, followed by `VALUES` and specific data values. The bottom section, titled "Output", shows the execution results for the first five statements, indicating that each insert operation was successful and affected one row.

Limit to 50000 rows

1 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2938', '1`

2 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2945', 'C`

3 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2952', 'E`

4 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2959', 'C`

5 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2966', 'E`

6 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2973', 'F`

7 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2980', 'C`

8 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2987', 'C`

9 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2994', 'E`

10 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3001', 'C`

11 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3008', 'L`

12 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3015', 'F`

13 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3022', 'C`

14 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3029', 'A`

15 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3036', 'A`

16 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3043', 'T`

17 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3050', 'F`

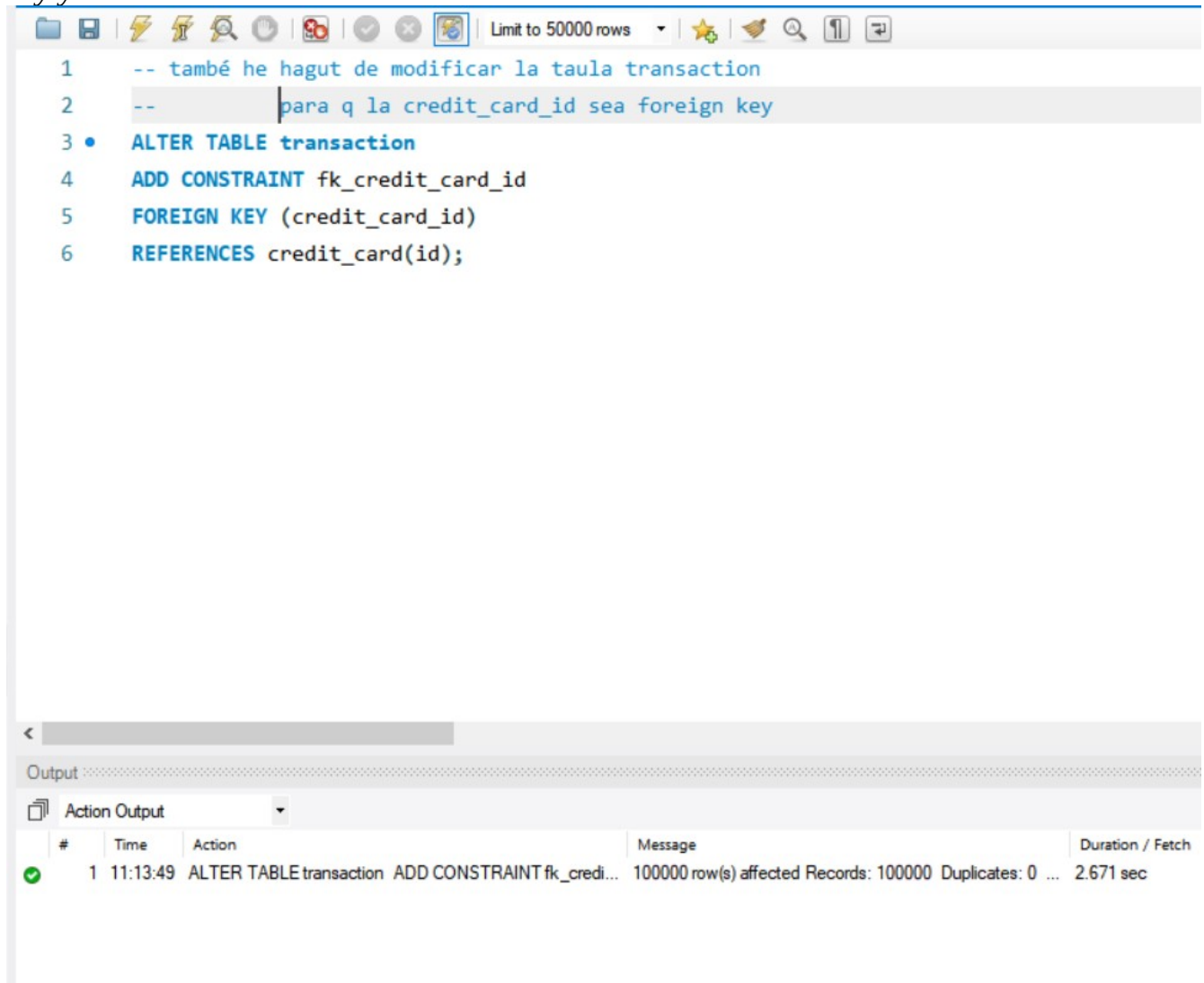
18 • `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3057', 'L`

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 4996	11:11:47	<code>INSERT INTO credit_card (id, iban, pan, pin, cvv, expir...</code>	1 row(s) affected	0.000 sec
✓ 4997	11:11:47	<code>INSERT INTO credit_card (id, iban, pan, pin, cvv, expir...</code>	1 row(s) affected	0.000 sec
✓ 4998	11:11:47	<code>INSERT INTO credit_card (id, iban, pan, pin, cvv, expir...</code>	1 row(s) affected	0.016 sec
✓ 4999	11:11:47	<code>INSERT INTO credit_card (id, iban, pan, pin, cvv, expir...</code>	1 row(s) affected	0.000 sec
✓ 5000	11:11:47	<code>INSERT INTO credit_card (id, iban, pan, pin, cvv, expir...</code>	1 row(s) affected	0.016 sec

-També he hagut de modificar la taula transaction para q la credit_card_id sea foreign key y tener así las tablas relacionadas entre si.



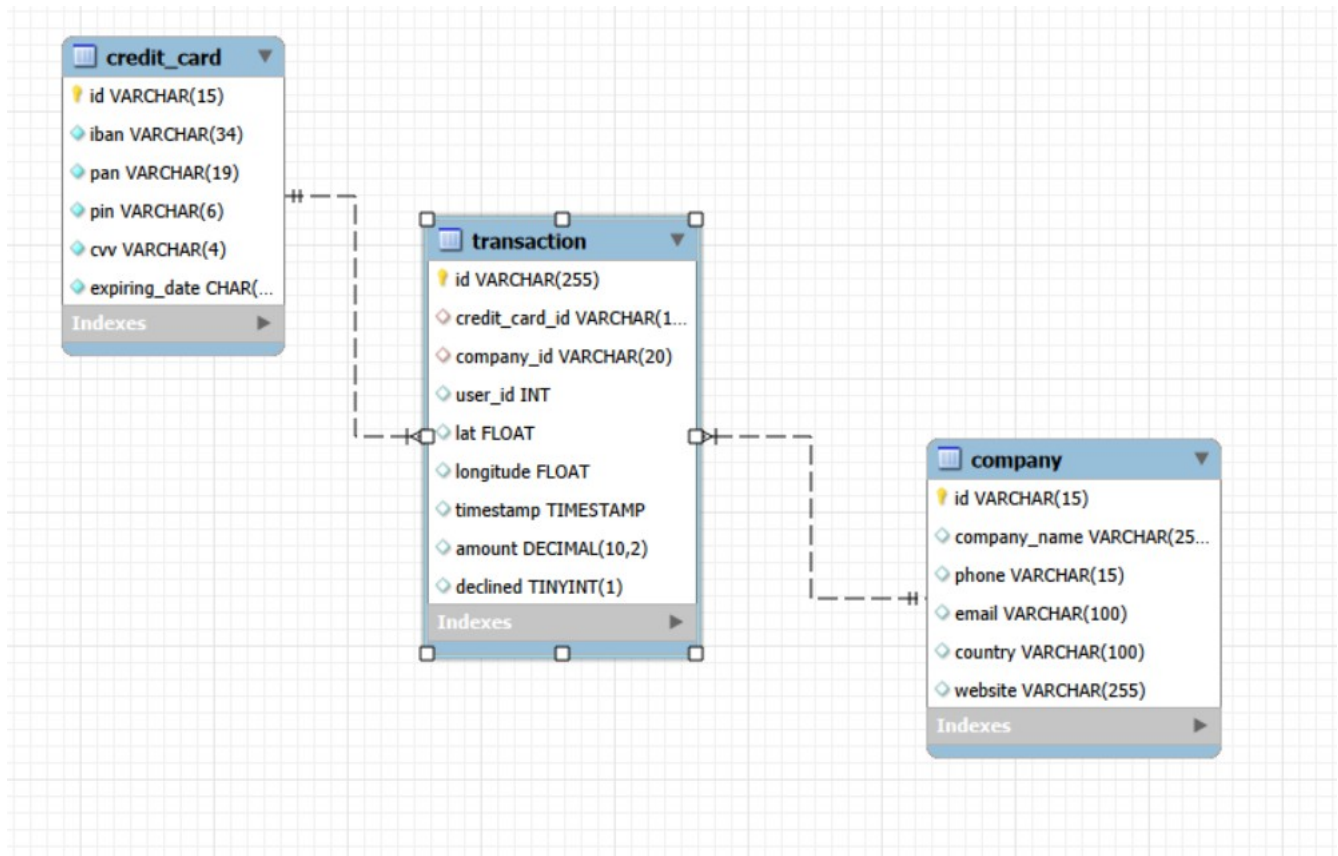
The screenshot shows a database management interface with a SQL editor and an output window. The SQL editor contains the following code:

```
1  -- també he hagut de modificar la taula transaction
2  -- para q la credit_card_id sea foreign key
3  • ALTER TABLE transaction
4  ADD CONSTRAINT fk_credit_card_id
5  FOREIGN KEY (credit_card_id)
6  REFERENCES credit_card(id);
```

The output window shows the results of the SQL execution:

#	Time	Action	Message	Duration / Fetch
1	11:13:49	ALTER TABLE transaction ADD CONSTRAINT fk_cred...	100000 row(s) affected Records: 100000 Duplicates: 0 ...	2.671 sec

Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.



Base de datos Transactions:

tabla de hechos: transaction (PK id)

tablas dimensión: credit_card(PK id) y company(PK id)

Las PK de las tablas de dimensión son FK de la tabla de hechos.

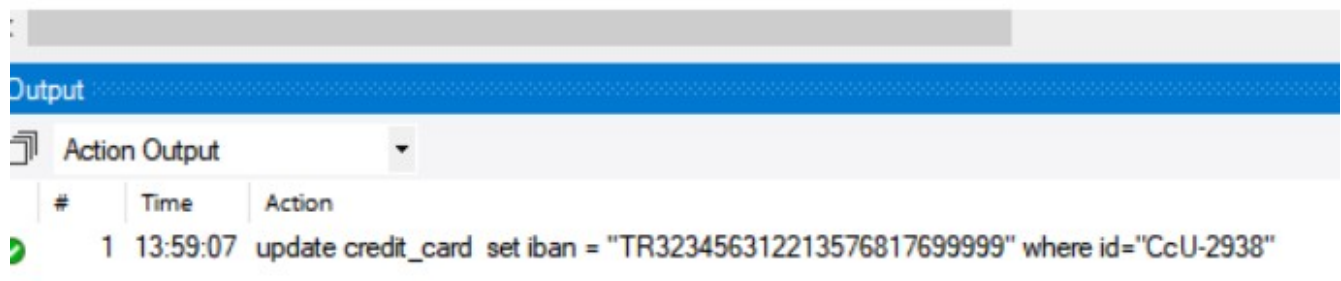
Exercici 5

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.



The screenshot shows a SQL query editor with a toolbar at the top containing icons for file operations, execution, and search. The query text is as follows:

```
1 • update credit_card
2   set iban = "TR323456312213576817699999"
3   where id="CcU-2938";
```



The screenshot shows the 'Output' window with a dropdown menu set to 'Action Output'. It displays a single log entry for the executed query.

#	Time	Action
1	13:59:07	update credit_card set iban = "TR323456312213576817699999" where id="CcU-2938"

Exercici 6

En la taula "transaction" ingresa una nova transacció amb la següent informació:

Id ,108B1D1D-5B23-A76C-55EF-
C568E49A99DD ,credit_card_id ,CcU-9999 ,company_id ,b-9999 ,user_id ,9999 ,lat
,829.999 ,longitude ,-117.999 ,amount ,111.11 ,declined ,0

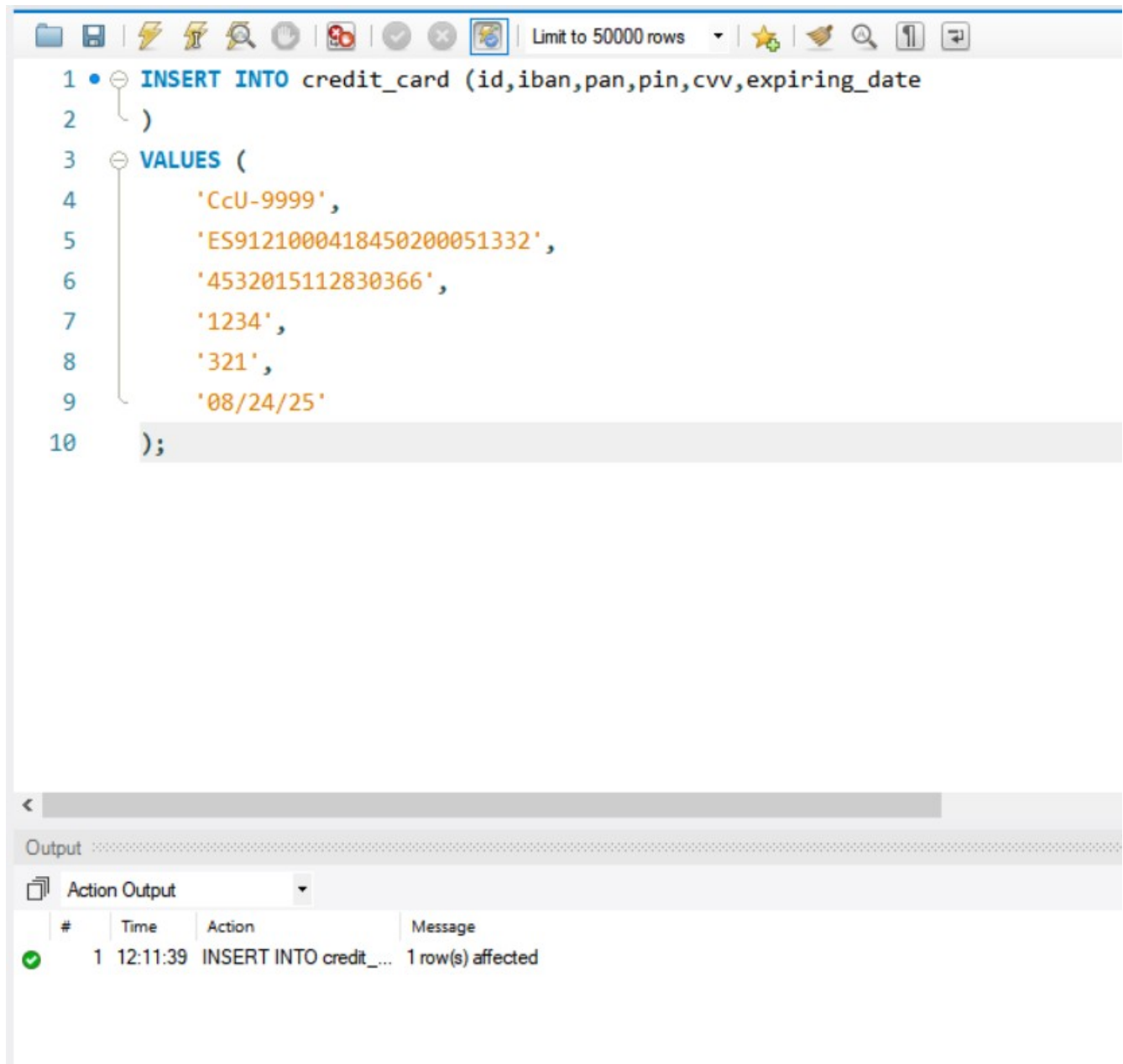
Al hacer un insert me da Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`transactions`.`transaction`, CONSTRAINT `transaction\_ibfk\_1` FOREIGN KEY (`company\_id`) REFERENCES `company` (`id`))

Le pedimos los datos de la compañía a RRHH para poder introducirla en la tabla de company y cumplir con la necesidad de clave foránea.

```
1 • INSERT INTO company ( id, company_name, phone, email, country, website
2   )
3   VALUES ( 'B-9999', 'Empresa Prueba S.L.', '+34931234567', 'contacto@empresaprueba.com',
4     'Spain', 'https://www.empresaprueba.com'
5   );
6
```

Output			
Action Output			
#	Time	Action	Message
✓ 1	12:04:43	INSERT INTO compa...	1 row(s) affected

Solucionamos el requerimiento de credit_card_id como clave foránea pidiendole los datos de la tarjeta a RRHH



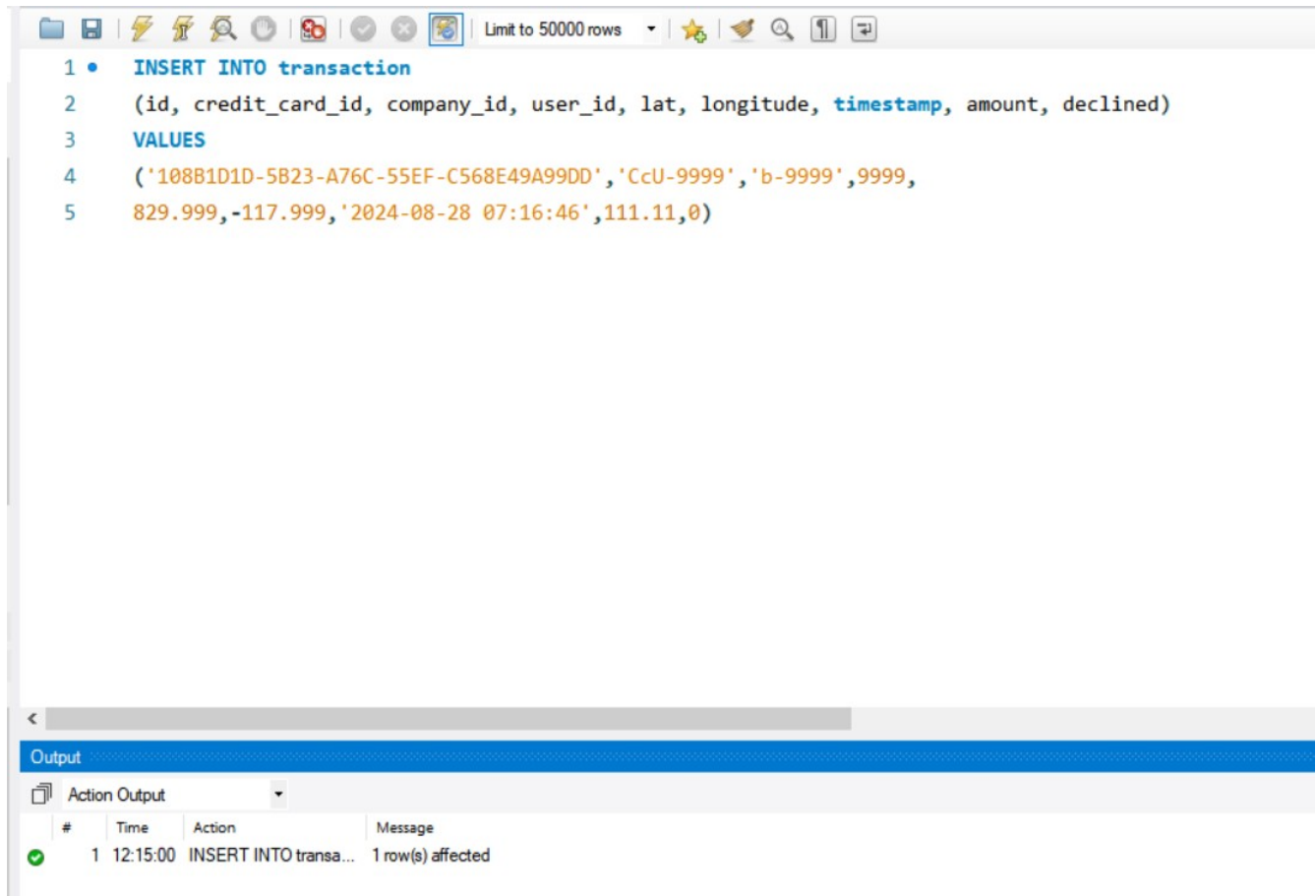
The screenshot displays a database management interface. The top toolbar includes icons for file operations, execution, and search, along with a 'Limit to 50000 rows' dropdown. The main editor shows an SQL statement:

```
1 • INSERT INTO credit_card (id,iban,pan,pin, cvv,expiring_date
2   )
3   VALUES (
4       'CcU-9999',
5       'ES9121000418450200051332',
6       '4532015112830366',
7       '1234',
8       '321',
9       '08/24/25'
10  );
```

Below the editor, the 'Output' tab is active, showing the 'Action Output' for the executed statement:

#	Time	Action	Message
1	12:11:39	INSERT INTO credit_...	1 row(s) affected

Intentamos crear la transacción de nuevo (RRHH también nos facilita la hora(timestamp) de la transacción 2024-08-28 07:16:46)



The screenshot displays a database client window with a toolbar at the top. The main area contains an SQL `INSERT` statement for a table named `transaction`. The statement includes column names and their corresponding values, with the timestamp set to '2024-08-28 07:16:46'. Below the SQL editor, the 'Output' tab is active, showing a table with the execution results. The table has four columns: '#', 'Time', 'Action', and 'Message'. A single row indicates that the `INSERT` operation was successful, affecting 1 row.

```
1 • INSERT INTO transaction
2   (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined)
3   VALUES
4   ('108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', 9999,
5   829.999, -117.999, '2024-08-28 07:16:46', 111.11, 0)
```

#	Time	Action	Message
✓ 1	12:15:00	INSERT INTO transa...	1 row(s) affected

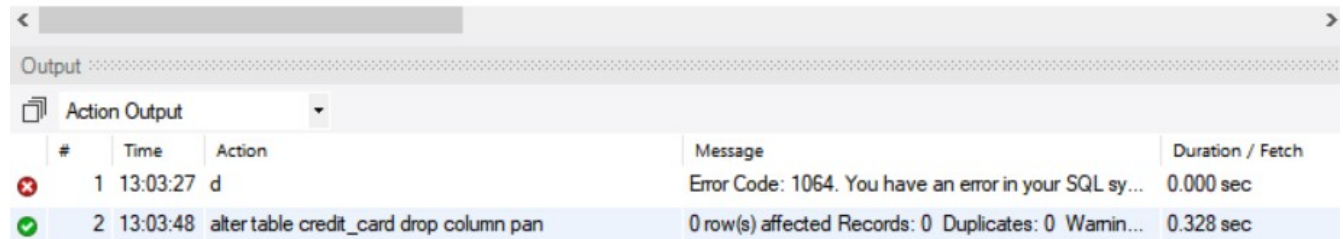
Exercici 7

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recordar mostrar el canvi realitzat.



The screenshot shows a SQL editor window with multiple tabs. The active tab is 'SQL File 17*'. The script contains the following SQL commands:

```
1  -- Exercici 7
2  -- Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula cred
3  -- Recordar mostrar el canvi realitzat.
4  • alter table credit_card drop column pan;
```



The screenshot shows the 'Output' window with the 'Action Output' tab selected. It displays a log of SQL execution actions.

#	Time	Action	Message	Duration / Fetch
1	13:03:27	d	Error Code: 1064. You have an error in your SQL sy...	0.000 sec
2	13:03:48	alter table credit_card drop column pan	0 row(s) affected Records: 0 Duplicates: 0 Wamin...	0.328 sec

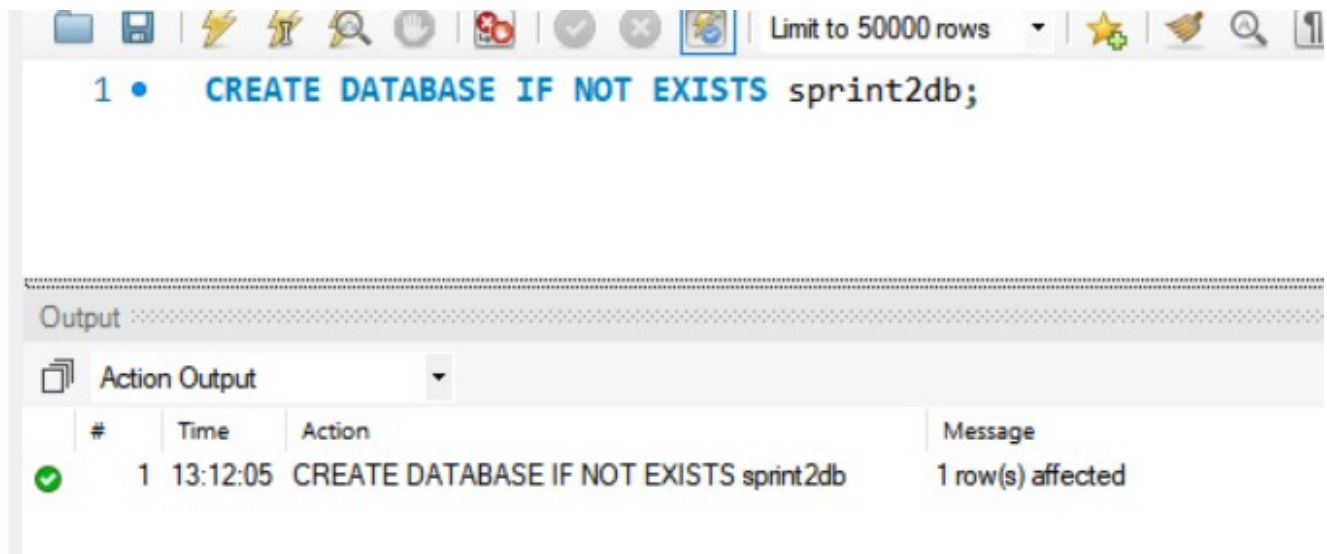
Exercici 8

Descarrega els arxius CSV que trobaràs a l'apartat de recursos:

- american_users.csv
- european_users.csv
- companies.csv
- credit_cards.csv
- transactions.csv

Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:

La taula de products.csv l'utilitzarem més endavant.



Creo les taules users euro i users america per després unir-les en una mateixa taula users_all.

tabla users_america

```
21 • CREATE TABLE IF NOT EXISTS users_america (  
22     id VARCHAR(15) PRIMARY KEY,  
23     name VARCHAR(50),  
24     surname VARCHAR(100),  
25     phone VARCHAR(25),  
26     email VARCHAR(255),  
27     birth_date DATE,  
28     country VARCHAR(100),  
29     city VARCHAR(100),  
30     postal_code VARCHAR(15),  
31     address VARCHAR(255),  
32     signup_date DATE,  
33     user_segment VARCHAR(50),  
34     income_band VARCHAR(20)  
35 );  
36
```

<

Output



Action Output



#	Time	Action	Message
✓ 1	11:54:12	CREATE TABLE IF NOT EXISTS users_america (...	0 row(s) affected

users_europe

```
21 • CREATE TABLE IF NOT EXISTS users_europe (  
22     id VARCHAR(15) PRIMARY KEY,  
23     name VARCHAR(50),  
24     surname VARCHAR(100),  
25     phone VARCHAR(25),  
26     email VARCHAR(255),  
27     birth_date DATE,  
28     country VARCHAR(100),  
29     city VARCHAR(100),  
30     postal_code VARCHAR(15),  
31     address VARCHAR(255),  
32     signup_date DATE,  
33     user_segment VARCHAR(50),  
34     income_band VARCHAR(20)  
35 );  
36
```

Output			
Action Output			
#	Time	Action	Message
✓ 1	11:55:36	CREATE TABLE IF NOT EXISTS users_europe (...	0 row(s) affected

Relleno las tablas users euro y america con los datos de sus csv.

```
14  
15 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__american_users.csv'  
16 INTO TABLE american_users  
17 FIELDS TERMINATED BY ';'   
18 OPTIONALLY ENCLOSED BY ''''  
19 LINES TERMINATED BY '\n'  
20 IGNORE 1 LINES;  
21  
22 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv'  
23 INTO TABLE european_users  
24 FIELDS TERMINATED BY ';'   
25 OPTIONALLY ENCLOSED BY ''''  
26 LINES TERMINATED BY '\n'  
27 IGNORE 1 LINES;  
28
```

Modifico formato clumnas birth_date

La columna birth_date no tienen un formato adecuado para una variable Date así que cambio las columnas de birthdate a texto para luego hacer la conversion a Date con STR_TO_DATE(). Esto lo hice pensando en hacer cálculos con las fechas aunque al final del ejercicio comprobé que no habría sido necesario ya que no nos pidieron cálculo alguno

```
5
6 • alter table users_america modify column birth_date text;
7 • alter table users_europe modify column birth_date text;
8
```

Output

Action Output

	#	Time	Action	Message	Duration / Fetch
✓	1	12:29:34	alter table users_america modify column birth_date t...	0 row(s) affected Records: 0 Duplicates: 0 Wamin...	0.109 sec
✓	2	12:29:34	alter table users_europe modify column birth_date text	0 row(s) affected Records: 0 Duplicates: 0 Wamin...	0.094 sec

Creo columna nueva provisional birth_date_DATE formato DATE para pegar las fechas despues de convertirlas a DATE desde birth_date

```
18
19 • ALTER TABLE users_america
20   ADD COLUMN birth_date_DATE date;
21 • ALTER TABLE users_europe
22   ADD COLUMN birth_date_DATE date;
23
```

```
24 • update users_america
25   set birth_date_DATE = STR_TO_DATE(birth_date, '%b %d, %Y') ;
26   -- select * from users_america limit 1;
27 • update users_europe
28   set birth_date_DATE = STR_TO_DATE(birth_date, '%b %d, %Y') ;
```

Output



Action Output

#	Time	Action	Message
✓ 1	13:15:13	update users_america set birth_date_DATE = STR_TO_DATE(birth_date, '%b %d, %Y')	1009 row(s) affected R
✓ 2	13:15:37	update users_europe set birth_date_DATE = STR_TO_DATE(birth_date, '%b %d, %Y')	3990 row(s) affected R

Creo la tabla users_all donde juntaremos los usuarios de EEUU y Europa.

```
4 • CREATE TABLE IF NOT EXISTS users_all (  
5     id VARCHAR(15) PRIMARY KEY,  
6     name VARCHAR(50),  
7     surname VARCHAR(100),  
8     phone VARCHAR(25),  
9     email VARCHAR(255),  
10    birth_date DATE,  
11    country VARCHAR(100),  
12    city VARCHAR(100),  
13    postal_code VARCHAR(15),  
14    address VARCHAR(255),  
15    signup_date DATE,  
16    user_segment VARCHAR(50),  
17    income_band VARCHAR(20),  
18    region VARCHAR(50)  
19 );
```



Output



Action Output

#	Time	Action	Message
✓ 1	11:41:53	CREATE TABLE IF NOT EXISTS users_all (id V...	0 row(s) affected

Elimino las cols con el formato original de fecha

Cambio el nombre de la nueva columna con formato DATE al nombre de la columna q eliminé en el paso anterior

Ordeno las columnas de users europe y america para que coincidan con users all y poder juntarlas más fácilmente

Copiar la tabla users america en users all rellenando field region america. Los insert se hacen a través de una sub consulta

```
29
30 • alter table users_america drop column birth_date;
31 • alter table users_europe drop column birth_date;
32 • alter table users_america rename column birth_date_DATE to birth_date;
33 • alter table users_europe rename column birth_date_DATE to birth_date;
34
35
36 • ALTER TABLE users_america MODIFY birth_date date AFTER email;
37 • ALTER TABLE users_europe MODIFY birth_date date AFTER email;
38
39 • INSERT INTO users_all (
40     id,name,surname,phone,email,birth_date,country,
41     city,postal_code,address,signup_date,user_segment,income_band,region
42 )
43 SELECT
44     id,name,surname,phone,email,birth_date,country,
45     city,postal_code,address,signup_date,user_segment,income_band,
46     'America'
47 FROM users_america;
48
49 • INSERT INTO users_all (
50     id,name,surname,phone,email,birth_date,country,
51     city,postal_code,address,signup_date,user_segment,income_band,region
52 )
53 SELECT
54     id,name,surname,phone,email,birth_date,country,
55     city,postal_code,address,signup_date,user_segment,income_band,
56     'Europe'
57 FROM users_europe;
```

Output				
Action Output				
#	Time	Action	Message	
9	13:23:27	alter table users_europe rename column birth_date_DATE to birth_date	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	
10	13:32:05	ALTER TABLE users_america MODIFY birth_date date AFTER email	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	
11	13:32:22	ALTER TABLE users_europe MODIFY birth_date date AFTER email	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	
12	13:36:18	INSERT INTO users_all (id,name,surname,phone,email,birth_date,country, city,postal_code,...	1010 row(s) affected Records: 1010 Duplicates: 0 Warnings: 0	
13	13:37:01	INSERT INTO users_all (id,name,surname,phone,email,birth_date,country, city,postal_code,...	3990 row(s) affected Records: 3990 Duplicates: 0 Warnings: 0	

Creo la tabla transaction que, entre otras, estará relacionada con users_all para poder contar las transacciones por usuario.

```
91 • CREATE TABLE IF NOT EXISTS transaction (
92     id VARCHAR(255) PRIMARY KEY,
93     card_id VARCHAR(15),
94     business_id VARCHAR(20),
95     product_ids INT,
96     timestamp TIMESTAMP,
97     amount DECIMAL(10, 2),
98     declined BOOLEAN,
99     user_id VARCHAR(15),
100     lat FLOAT,
101     longitude FLOAT,
102     discount_amount DECIMAL(10,2),
103     tax_amount DECIMAL(10,2),
104     shipping_amount DECIMAL(10,2),
105     channel VARCHAR(20),
106     campaign_id VARCHAR(50),
107     device_type VARCHAR(20),
108     is_international BOOLEAN,
109     decline_reason VARCHAR(100),
110     distance_km DECIMAL(10,2),
111     FOREIGN KEY (card_id) REFERENCES credit_card(id),
112     FOREIGN KEY (business_id) REFERENCES company(id),
113     FOREIGN KEY (product_ids) REFERENCES products(id),
114     FOREIGN KEY (user_id) REFERENCES users_all(id)
115 );
116
117
```

Output

📄

Action Output

#	Time	Action	Message	Duration
✓ 4	14:11:38	CREATE TABLE IF NOT...	0 row(s) affected	0.046 sec
✓ 5	14:18:06	CREATE TABLE product...	0 row(s) affected	0.047 sec
✗ 6	14:19:10	CREATE TABLE IF NOT...	Error Code: 1824. Failed to open the referenced table 'credit_cards'	0.016 sec
✗ 7	14:19:45	CREATE TABLE IF NOT...	Error Code: 3780. Referencing column 'product_ids' and referenced column 'id' in foreign key ...	0.016 sec
✗ 8	14:20:32	CREATE TABLE IF NOT...	Error Code: 3780. Referencing column 'user_id' and referenced column 'id' in foreign key cons...	0.000 sec
✓ 9	14:21:21	CREATE TABLE IF NOT...	0 row(s) affected	0.171 sec

**los errores son de otras consultas previas*

```
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__transactions.csv'
INTO TABLE transaction
FIELDS TERMINATED BY ';'
OPTIONALLY ENCLOSED BY ''
LINES TERMINATED BY '\n'
IGNORE 1 LINES;
```

Exercici 9

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

```
146 • select id, name, surname from users_all
147   where id In (
148       select user_id from transaction
149       group by user_id
150       having count(user_id) > 80
151   );
```

```
152   -- 185  110
153   -- 289  94
154   -- 318  91
155   -- 454  81
```

<

Result Grid   Filter Rows: | Edit:    | Export/Import

	id	name	surname
▶	185	Molly	Gilliam
	289	Dxwqi	Hwcru
	318	Bnyr	Astuw
	454	Sfzzoh	Xqvfridxs
*	NULL	NULL	NULL

users_all 30 x

Output

 Action Output ▼

	#	Time	Action	Message
✓	1	20:15:35	select id, name, sum...	4 row(s) returned

Exercici 10

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

Primero creo la tabla credit_card y cargo el csv con los datos de las targetas de crédito en la tabla credit card

```
160 • CREATE TABLE IF NOT EXISTS credit_card (  
161     -- id,  
162     id VARCHAR(15) PRIMARY KEY,  
163     -- user_id,  
164     user_id VARCHAR(15) not NULL,  
165     -- iban,  
166     iban VARCHAR(34) NOT NULL,  
167     -- pan,  
168     pan VARCHAR(19) NOT NULL,  
169     -- pin,  
170     pin VARCHAR(6) NOT NULL,  
171     -- cvv,  
172     cvv VARCHAR(4) NOT NULL,  
173     -- track1,  
174     track1 VARCHAR(100),  
175     -- %B8383712448554646^WovsxejDpwiev^86041142?7,  
176     -- track2,  
177     track2 VARCHAR(100),  
178     -- %B7653863056044187=8007163336?3,  
179     -- expiring_date,  
180     expiring_date CHAR(8) NOT NULL, -- mm/dd/yy  
181     -- card_type,  
182     card_type VARCHAR(25) NOT NULL,  
183     -- card_renewal_flag  
184     card_renewal_flag BOOLEAN  
185 );  
186  
187 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__credit_cards.csv'  
188 INTO TABLE credit_card  
189 FIELDS TERMINATED BY ','  
190 OPTIONALLY ENCLOSED BY ''''  
191 LINES TERMINATED BY '\n'  
192 IGNORE 1 LINES; -- 5000 rows
```

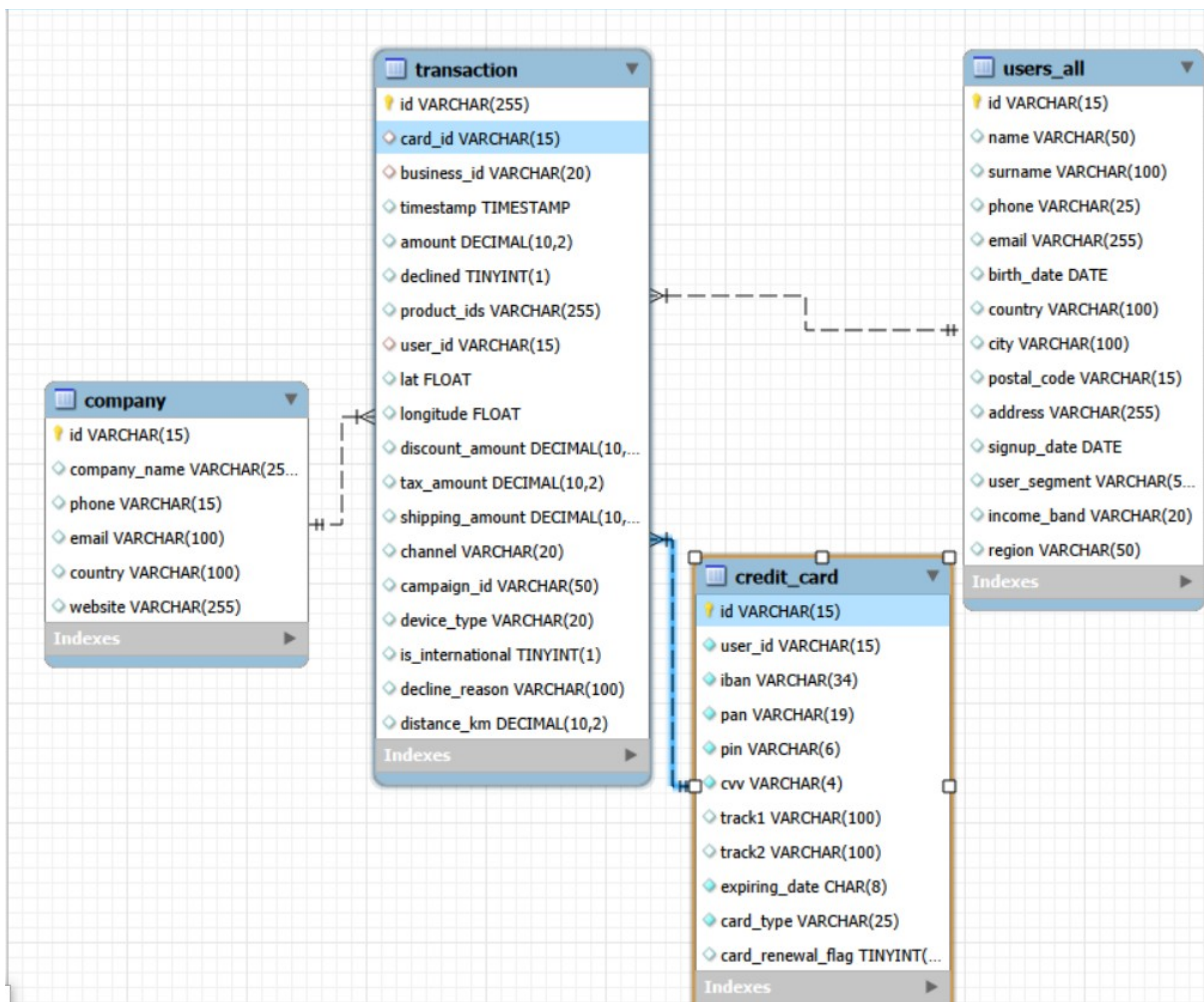
Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	21:00:36	LOAD DATA INFILE 'C:/Pr...	5000 row(s) affected Records: 5000 Deleted: 0 Skipped: 0 Warnings: 0	0.281 sec

Hago lo mismo con la tabla company y su csv.

```
209 • CREATE TABLE IF NOT EXISTS company (  
210     id VARCHAR(15) PRIMARY KEY,  
211     company_name VARCHAR(255),  
212     phone VARCHAR(15),  
213     email VARCHAR(100),  
214     country VARCHAR(100),  
215     website VARCHAR(255)  
216 );  
217  
218 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_companies.csv'  
219 INTO TABLE companies  
220 FIELDS TERMINATED BY ';'   
221 OPTIONALLY ENCLOSED BY ''   
222 LINES TERMINATED BY '\n'  
223 IGNORE 1 LINES;  
224
```



Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

```
222
223 -- Mostra la mitjana d'amount per IBAN de les targetes de crèdit
224 -- a la companyia Donec Ltd, utilitza almenys 2 taules.
225 • select cr.iban, avg(t.amount) as "AVG(iban)",co.company_name
226 from credit_card as cr -- 5000 iban en credit card
227 join transaction as t on t.user_id = cr.user_id
228 join company as co on t.business_id = co.id
229 where co.company_name = "Donec Ltd" -- 449 transaccions donec
230 group by cr.iban -- 371 ibans donec
231 ;
232
233
```

<

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	iban	AVG(iban)	company_name
▶	CA796226931885124489219768	13.910000	Donec Ltd
	US473611582698831054568995	371.700000	Donec Ltd
	CA418666139713972517702166	310.040000	Donec Ltd
	CA472858869714205818784199	339.725000	Donec Ltd

Result 17 x

Output

 Action Output

#	Time	Action	Message
✓ 1	10:51:33	select cr.iban, avg(t.amount) as "AVG(iban)",co.company_...	371 row(s) returned

Nivell 2

Exercici 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

```
236 -- Identifica els cinc dies que es va generar la quantitat més gran
237 -- d'ingressos a l'empresa per vendes.
238 -- Mostra la data de cada transacció juntament amb el total de les vendes.
239 • select avg(amount), date(timestamp) from transaction
240 group by date(timestamp)
241 order by avg(amount) desc
242 limit 5
243 ;
244 |
```

< **Result Grid** | | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows: ☐

	avg(amount)	date(timestamp)
▶	449.204286	2018-11-11
	436.837778	2016-03-16
	435.985833	2020-10-25
	429.122692	2020-09-26
	411.701818	2024-02-11

Result 6 x

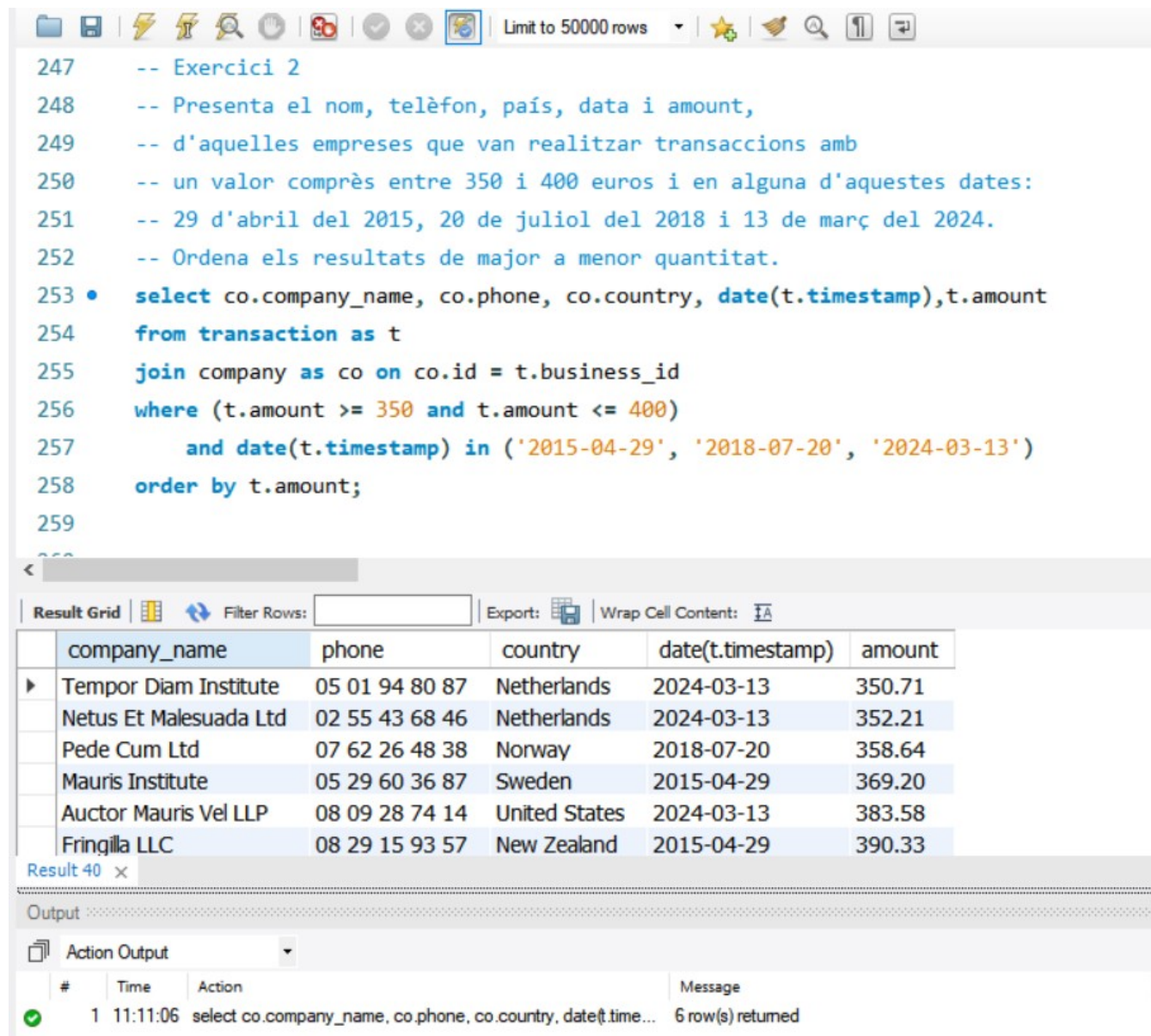
Output

Action Output

#	Time	Action	Message	Duration
✓ 1	10:31:40	select avg(amount), date(timestamp) from transacti...	5 row(s) returned	0.157 sec

Exercici 2

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.



The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, search, and execution, along with a 'Limit to 50000 rows' dropdown. The SQL editor contains the following query:

```
247 -- Exercici 2
248 -- Presenta el nom, telèfon, país, data i amount,
249 -- d'aquelles empreses que van realitzar transaccions amb
250 -- un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates:
251 -- 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024.
252 -- Ordena els resultats de major a menor quantitat.
253 • select co.company_name, co.phone, co.country, date(t.timestamp),t.amount
254 from transaction as t
255 join company as co on co.id = t.business_id
256 where (t.amount >= 350 and t.amount <= 400)
257 and date(t.timestamp) in ('2015-04-29', '2018-07-20', '2024-03-13')
258 order by t.amount;
259
```

Below the editor, the 'Result Grid' shows the query results. The table has 6 columns: company_name, phone, country, date(t.timestamp), and amount. The results are sorted by amount in descending order.

	company_name	phone	country	date(t.timestamp)	amount
▶	Tempor Diam Institute	05 01 94 80 87	Netherlands	2024-03-13	350.71
	Netus Et Malesuada Ltd	02 55 43 68 46	Netherlands	2024-03-13	352.21
	Pede Cum Ltd	07 62 26 48 38	Norway	2018-07-20	358.64
	Mauris Institute	05 29 60 36 87	Sweden	2015-04-29	369.20
	Auctor Mauris Vel LLP	08 09 28 74 14	United States	2024-03-13	383.58
	Fringilla LLC	08 29 15 93 57	New Zealand	2015-04-29	390.33

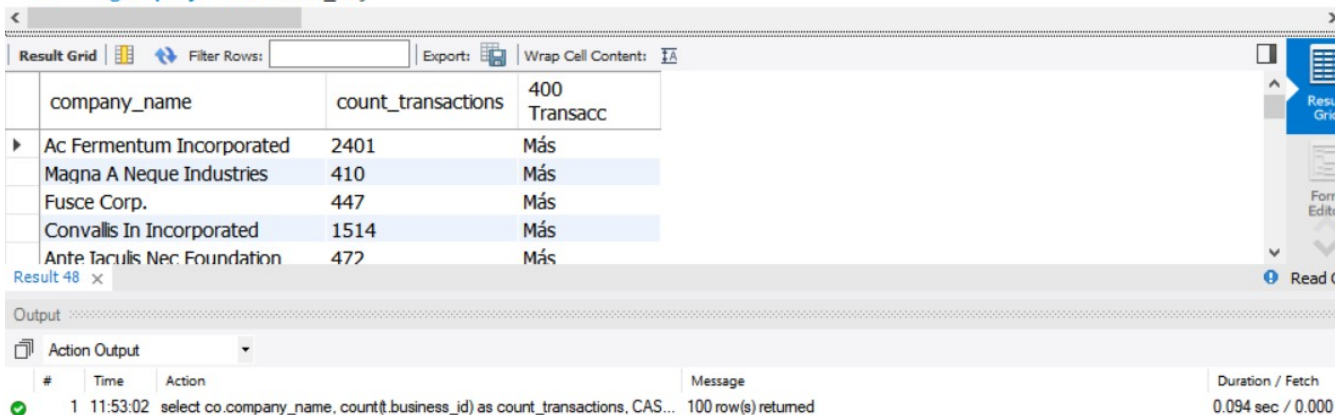
Below the result grid, the 'Output' section shows the 'Action Output' for the query execution. It indicates that the query returned 6 rows.

#	Time	Action	Message
1	11:11:06	select co.company_name, co.phone, co.country, date(t.time...	6 row(s) returned

Exercici 3

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.

```
262 -- Exercici 3
263 -- Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi,
264 -- la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent
265 -- les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.
266 • select co.company_name, count(t.business_id) as count_transactions,
267 CASE
268     WHEN count(t.business_id) > 400 THEN "Más"
269     WHEN count(t.business_id) = 400 THEN "Igual"
270     ELSE "Menos"
271 END as "400 Transacc"
272 from transaction as t
273 join company as co on co.id = t.business_id
274 group by t.business_id;
```



company_name	count_transactions	
	400	Transacc
Ac Fermentum Incorporated	2401	Más
Magna A Neque Industries	410	Más
Fusce Corp.	447	Más
Convalis In Incorporated	1514	Más
Ante Taculis Nec Foundation	472	Más

#	Time	Action	Message	Duration / Fetch
1	11:53:02	select co.company_name, count(t.business_id) as count_transactions, CAS...	100 row(s) returned	0.094 sec / 0.000

Exercici 4

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

```
277 -- Exercici 4
278 -- Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD
279 -- de la base de dades.
280 • select * FROM transaction
281 WHERE id = "000447FE-B650-4DCF-85DE-C7ED0EE1CAAD";
282 • DELETE FROM transaction
283 WHERE id = "000447FE-B650-4DCF-85DE-C7ED0EE1CAAD";
284
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	12:00:11	select * FROM transaction WHERE id = "000447FE-B650-4DCF-85DE-C7...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 2	12:00:26	DELETE FROM transaction WHERE id = "000447FE-B650-4DCF-85DE-C...	1 row(s) affected	0.031 sec

Exercici 5

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una **vista** que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.

```
291 -- Exercici 5
292 • create view VistaMarketing as
293     select co.company_name, co.phone, co.country, round(avg(t.amount), 2) as AVG_amount
294     from transaction as t
295     join company as co on t.business_id = co.id
296     group by t.business_id
297     order by avg_amount desc;
298 • select * from VistaMarketing;
299
```

	company_name	phone	country	AVG_amount
▶	Ac Fermentum Incorporated	06 85 56 52 33	Germany	330.77
	Ut Semper Foundation	01 60 36 33 06	Sweden	307.37
	Orci Adipiscing Limited	03 18 00 77 81	United Kingdom	306.27
	Dolor Vitae Limited	06 53 60 43 60	France	305.95
	Nunc In Foundation	03 79 81 18 92	Italy	305.40
	Maecenas Malesuada Fringilla Inc.	09 38 53 76 61	Netherlands	305.15
	Aliquet Sem Limited	03 75 57 02 01	Netherlands	304.21

VistaMarketing 57 x

Output

Action Output

#	Time	Action	Message
✓ 1	12:33:52	create view VistaMarketing as select co.company_name, co.phone, co.co...	0 row(s) affected
✓ 2	12:33:52	select * from VistaMarketing LIMIT 0, 50000	100 row(s) returned

Nivell 3

Exercici 1

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon:

```
318 • select card_id, sum(declined) as numDeclined
319 , CASE
320   WHEN (sum(declined) < 3) THEN "ACTIVE"
321   ELSE "INACTIVE"
322 END as "STATE CARD"
323 from (
324   SELECT card_id, date(timestamp) ,
325         ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY TIMESTAMP DESC) AS row_num, declined
326   FROM transaction
327   ) as sub
328 where row_num <= 3
329 group by card_id
330 order by sum(declined) desc;
331
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	card_id	numDeclined	STATE CARD
▶	CcS-9406	2	ACTIVE
	CcS-9473	1	ACTIVE
	CcS-9500	1	ACTIVE
	CcU-3113	1	ACTIVE

Result 11 x

Output

Action Output

#	Time	Action	Message
✓ 1	11:07:31	select card_id, sum(declined) as numDeclined , CASE WHEN (sum(declined) < 3) THEN "ACTIVE" ELS...	5000 row(s) returned

👉 Quantes targetes estan actives?

TOTES

Exercici 2

Crea una taula amb la qual puguem unir les dades de l'arxiu de products.csv amb la base de dades creada (ja que fins ara no podíem fer-ho), tenint en compte que des de transaction tens product_ids. Genera la següent consulta:

👉 Necessitem conèixer el nombre de vegades que s'ha venut cada producte.

Recursos

Objectius

- **Construir una base de dades relacional senzilla amb MySQL.**
- **Realitzar consultes únicament amb JOIN..**
- **Crear subconsultes SQL per a mostrar informació sense l'ús de JOIN.**

